Optimizing Customer Lifetime Value

Data-Driven Shopping Behavior Analysis

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The Bottom Line

Objective

Analyze 3,900 customers for revenue opportunities, inventory optimization, retention

Core Insight

Active customers, low subscription conversion, promotional redundancies

Key Action

Shift budget to subscription upsells, optimize inventory for high-margin categories

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Data Engineering Foundation

Robust Pipeline

End-to-end data pipeline ensuring 100% accuracy from raw CSV to SQL database

Data Cleaning

Python/Pandas features: age_group, purchase_frequency. Imputed missing ratings via category medians

01

Anomaly Detection

100% correlation between discount_applied and promo_code_used

02

Data Optimization

Dropped redundant data to clean reporting model

03

Advanced SQL

CTEs and Window Functions for precise segmentation

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Who is Our True Customer?

68%

Male Dominance

2,652 male customers out of 3,900 total

$157K

Male Revenue

Total revenue generated by male customers

$75K

Female Revenue

Total revenue generated by female customers

$62K

Young Adult

Highest revenue-generating demographic segment

Strategic Takeaway: Hyper-target campaigns on men in Young Adult bracket

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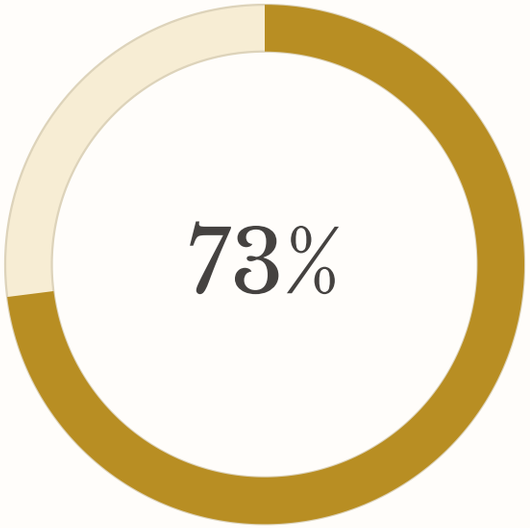
The Subscription Multiplier



Unsubscribed

2,847 customers remain unsubscribed

Repeat Buyers

Frequent non-subscribed customers

AOV Reality

Non-subscribers: $59.87

Subscribers: $59.49

Conversion Opportunity

Massive pool of frequent buyers need subscription offers

Strategic Takeaway: Aggressively market subscriptions to 2,518 repeat buyers, introduce subscriber-only upsells

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Product Performance

Jewelry

171 orders

Blouses

171 orders

Sandals

160 orders

Jackets

163 orders

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Quality & Discount Strategy

Top-Rated Products

Gloves: 3.86 avg rating

Sandals: 3.84 avg rating

Feature these to drive trust

Discount Efficiency

Discount users spent above overall average

Discounts increase cart size effectively

Redundancy Issue

100% overlap between promo codes and discounts

Stop separate campaigns immediately

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Actionable Recommendations

Fix Marketing Redundancy

Merge promo codes and general discounts. Stop wasting ad tracking resources on duplicate metrics

Launch Loyalty Pipeline

Target returning segment with aggressive subscription offers. Implement exclusive subscriber-tier bundles to push AOV past $60

Reallocate Inventory Budget

Double down on Clothing and Accessories. Fast-track Express shipping—customers spend $60.48 vs Standard's $58.46

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Strategic Priorities

Immediate

Merge promo codes and discounts

High Impact

Launch subscription pipeline for 2,518 repeat buyers

Quick Win

Double down on Clothing and Accessories inventory

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We Are Leaving Money on the Table

Highly active customer base with low subscription conversion and promotional redundancies

Target

Young Adult males

Convert

2,518 repeat buyers

Grow

Recurring revenue

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